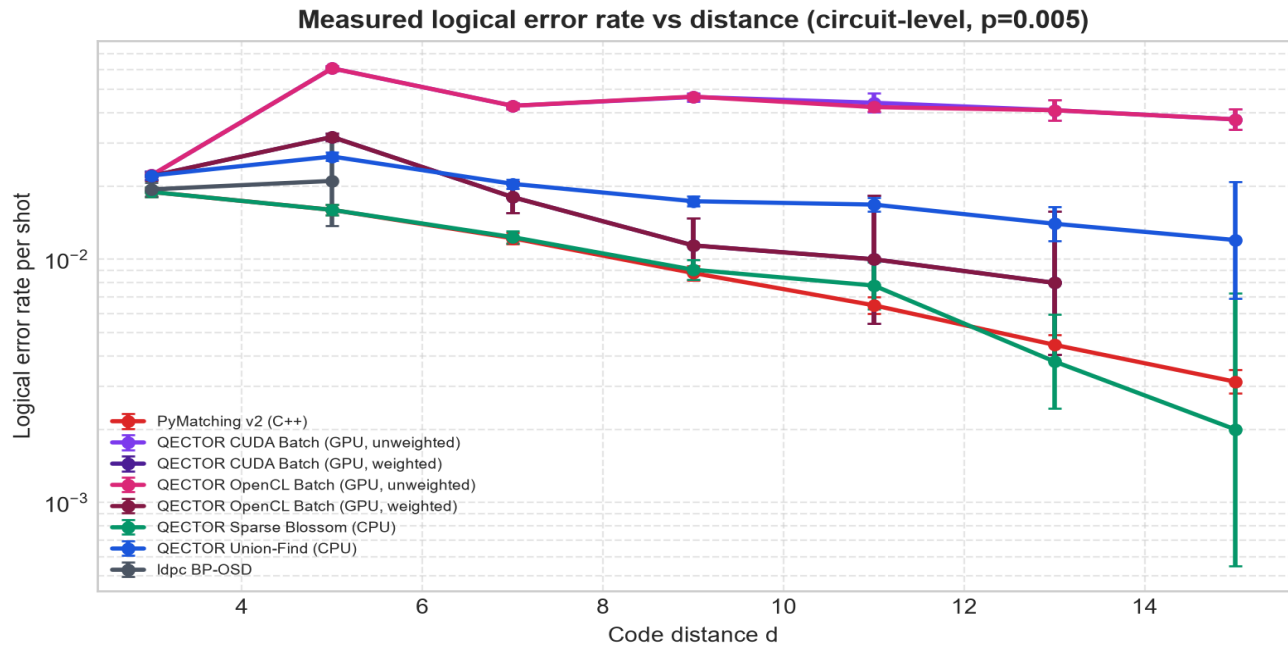
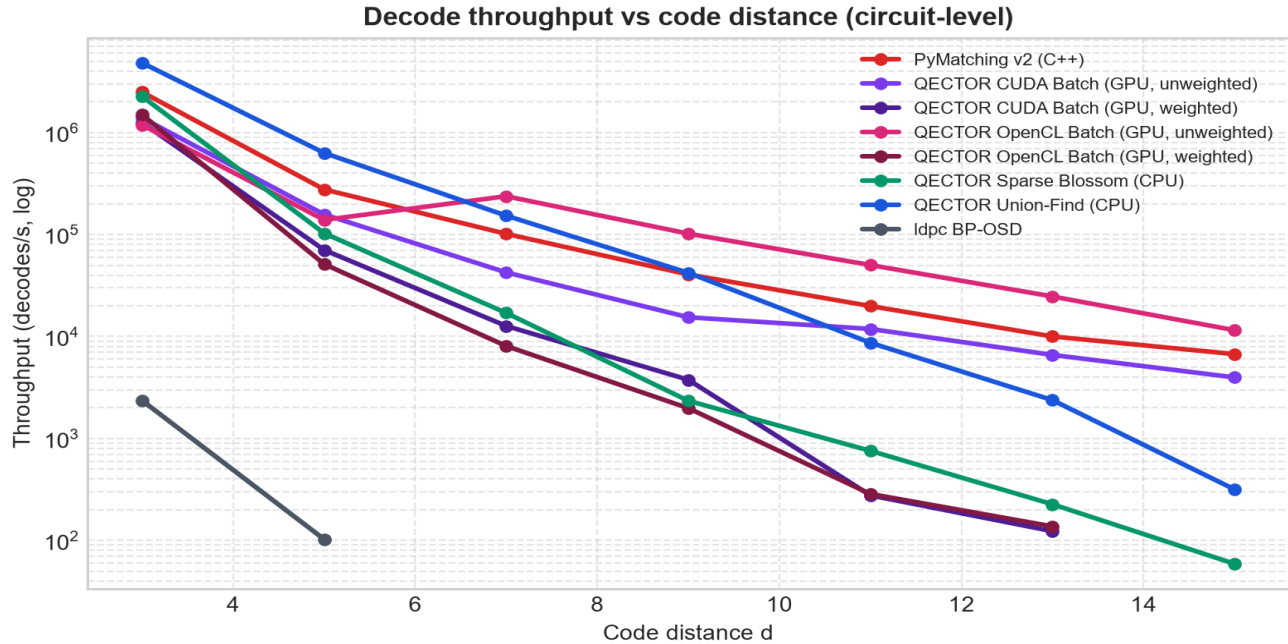


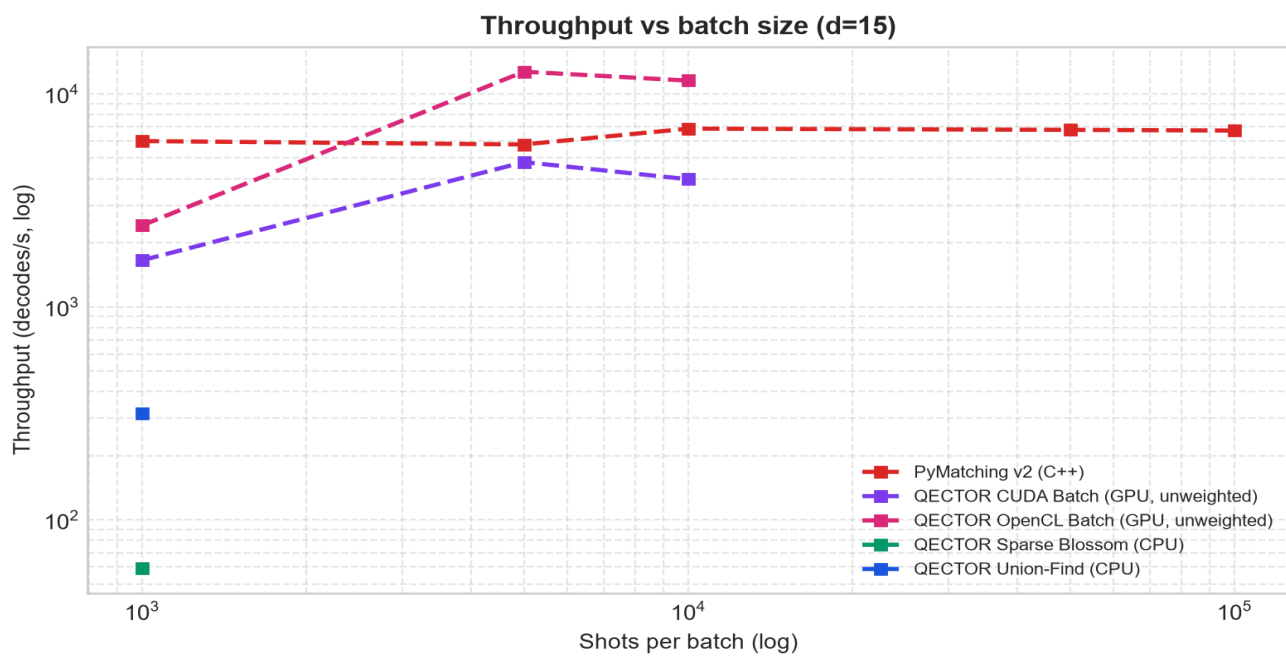
QECTOR v0.7.0 — circuit-level decoder comparison

Measured with `ler.estimate_ler_circuit_level`: identical circuit, DEM, samples and `decode_batch` resolver for every decoder, scored against the circuit's logical observables. Validated by `ler.assert_comparable`. No extrapolated cells.

Noise model: circuit-level, $p=0.005$ | Commit: 07966e4da5 | Generated: 2026-08-01T03:40:26Z

Platform: Windows-10-10.0.26100-SP0 | Python: 3.11.9





Measured results (185 cells)

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
PyMatching v2 (C++)	3	1,000	13	0.01300	[0.0076, 0.0221]	2,076,843	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	1,219,066	0.59x
QECTOR CUDA Batch (GPU, weighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	1,587,554	0.76x
QECTOR OpenCL Batch (GPU, unweighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	441,267	0.21x
QECTOR OpenCL Batch (GPU, weighted)	3	1,000	16	0.01600	[0.0099, 0.0258]	358,809	0.17x
QECTOR Sparse Blossom (CPU)	3	1,000	13	0.01300	[0.0076, 0.0221]	728,597	0.35x
QECTOR Union-Find (CPU)	3	1,000	16	0.01600	[0.0099, 0.0258]	887,863	0.43x
Idpc BP-OSD	3	1,000	12	0.01200	[0.0069, 0.0209]	2,435	0.001x
PyMatching v2 (C++)	3	5,000	84	0.01680	[0.0136, 0.0208]	2,371,242	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,670,565	0.70x
QECTOR CUDA Batch (GPU, weighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,369,638	0.58x
QECTOR OpenCL Batch (GPU, unweighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,518,188	0.64x
QECTOR OpenCL Batch (GPU, weighted)	3	5,000	105	0.02100	[0.0174, 0.0254]	1,189,343	0.50x
QECTOR Sparse Blossom (CPU)	3	5,000	84	0.01680	[0.0136, 0.0208]	1,491,335	0.63x
QECTOR Union-Find (CPU)	3	5,000	105	0.02100	[0.0174, 0.0254]	2,458,694	1.04x
Idpc BP-OSD	3	5,000	87	0.01740	[0.0141, 0.0214]	2,360	0.001x
PyMatching v2 (C++)	3	10,000	202	0.02020	[0.0176, 0.0231]	2,436,825	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	10,000	235	0.02350	[0.0207, 0.0267]	1,802,809	0.74x
QECTOR CUDA Batch (GPU, weighted)	3	10,000	238	0.02380	[0.0210, 0.0270]	1,408,867	0.58x
QECTOR OpenCL Batch (GPU, unweighted)	3	10,000	235	0.02350	[0.0207, 0.0267]	1,602,128	0.66x
QECTOR OpenCL Batch (GPU, weighted)	3	10,000	238	0.02380	[0.0210, 0.0270]	1,432,008	0.59x
QECTOR Sparse Blossom (CPU)	3	10,000	202	0.02020	[0.0176, 0.0231]	1,616,554	0.66x
QECTOR Union-Find (CPU)	3	10,000	233	0.02330	[0.0205, 0.0264]	2,809,225	1.15x
Idpc BP-OSD	3	10,000	199	0.01990	[0.0173, 0.0228]	2,371	0.001x
PyMatching v2 (C++)	3	50,000	960	0.01920	[0.0180, 0.0204]	2,501,288	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	50,000	1131	0.02262	[0.0214, 0.0240]	1,363,174	0.55x
QECTOR CUDA Batch (GPU, weighted)	3	50,000	1129	0.02258	[0.0213, 0.0239]	1,206,188	0.48x
QECTOR OpenCL Batch (GPU, unweighted)	3	50,000	1131	0.02262	[0.0214, 0.0240]	1,281,887	0.51x
QECTOR OpenCL Batch (GPU, weighted)	3	50,000	1129	0.02258	[0.0213, 0.0239]	1,328,900	0.53x
QECTOR Sparse Blossom (CPU)	3	50,000	960	0.01920	[0.0180, 0.0204]	2,174,518	0.87x
QECTOR Union-Find (CPU)	3	50,000	1128	0.02256	[0.0213, 0.0239]	4,510,274	1.80x
Idpc BP-OSD	3	50,000	969	0.01938	[0.0182, 0.0206]	2,356	0.001x
PyMatching v2 (C++)	3	100,000	1891	0.01891	[0.0181, 0.0198]	2,488,546	1.00x
QECTOR CUDA Batch (GPU, unweighted)	3	100,000	2215	0.02215	[0.0213, 0.0231]	1,391,074	0.56x
QECTOR CUDA Batch (GPU, weighted)	3	100,000	2201	0.02201	[0.0211, 0.0229]	1,252,127	0.50x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR OpenCL Batch (GPU, unweighted)	3	100,000	2215	0.02215	[0.0213, 0.0231]	1,186,666	0.48x
QECTOR OpenCL Batch (GPU, weighted)	3	100,000	2201	0.02201	[0.0211, 0.0229]	1,494,096	0.60x
QECTOR Sparse Blossom (CPU)	3	100,000	1891	0.01891	[0.0181, 0.0198]	2,259,953	0.91x
QECTOR Union-Find (CPU)	3	100,000	2210	0.02210	[0.0212, 0.0230]	4,816,445	1.94x
PyMatching v2 (C++)	5	1,000	19	0.01900	[0.0122, 0.0295]	273,688	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	1,000	60	0.06000	[0.0469, 0.0765]	86,112	0.32x
QECTOR CUDA Batch (GPU, weighted)	5	1,000	31	0.03100	[0.0219, 0.0437]	36,620	0.13x
QECTOR OpenCL Batch (GPU, unweighted)	5	1,000	60	0.06000	[0.0469, 0.0765]	123,236	0.45x
QECTOR OpenCL Batch (GPU, weighted)	5	1,000	31	0.03100	[0.0219, 0.0437]	28,726	0.10x
QECTOR Sparse Blossom (CPU)	5	1,000	19	0.01900	[0.0122, 0.0295]	97,117	0.35x
QECTOR Union-Find (CPU)	5	1,000	26	0.02600	[0.0178, 0.0378]	319,336	1.17x
ldpc BP-OSD	5	1,000	21	0.02100	[0.0138, 0.0319]	103	0x
PyMatching v2 (C++)	5	5,000	73	0.01460	[0.0116, 0.0183]	298,064	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	5,000	336	0.06720	[0.0606, 0.0745]	233,926	0.79x
QECTOR CUDA Batch (GPU, weighted)	5	5,000	181	0.03620	[0.0314, 0.0417]	106,413	0.36x
QECTOR OpenCL Batch (GPU, unweighted)	5	5,000	336	0.06720	[0.0606, 0.0745]	211,317	0.71x
QECTOR OpenCL Batch (GPU, weighted)	5	5,000	181	0.03620	[0.0314, 0.0417]	72,852	0.24x
QECTOR Sparse Blossom (CPU)	5	5,000	73	0.01460	[0.0116, 0.0183]	103,810	0.35x
QECTOR Union-Find (CPU)	5	5,000	148	0.02960	[0.0253, 0.0347]	526,643	1.77x
PyMatching v2 (C++)	5	10,000	166	0.01660	[0.0143, 0.0193]	302,401	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	10,000	600	0.06000	[0.0555, 0.0648]	244,247	0.81x
QECTOR CUDA Batch (GPU, weighted)	5	10,000	316	0.03160	[0.0283, 0.0352]	90,122	0.30x
QECTOR OpenCL Batch (GPU, unweighted)	5	10,000	600	0.06000	[0.0555, 0.0648]	185,942	0.61x
QECTOR OpenCL Batch (GPU, weighted)	5	10,000	316	0.03160	[0.0283, 0.0352]	58,504	0.19x
QECTOR Sparse Blossom (CPU)	5	10,000	166	0.01660	[0.0143, 0.0193]	113,171	0.37x
QECTOR Union-Find (CPU)	5	10,000	268	0.02680	[0.0238, 0.0302]	529,235	1.75x
PyMatching v2 (C++)	5	50,000	809	0.01618	[0.0151, 0.0173]	305,102	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	50,000	3041	0.06082	[0.0588, 0.0629]	151,108	0.49x
QECTOR CUDA Batch (GPU, weighted)	5	50,000	1561	0.03122	[0.0297, 0.0328]	68,196	0.22x
QECTOR OpenCL Batch (GPU, unweighted)	5	50,000	3041	0.06082	[0.0588, 0.0629]	137,818	0.45x
QECTOR OpenCL Batch (GPU, weighted)	5	50,000	1561	0.03122	[0.0297, 0.0328]	49,724	0.16x
QECTOR Sparse Blossom (CPU)	5	50,000	810	0.01620	[0.0151, 0.0173]	101,141	0.33x
QECTOR Union-Find (CPU)	5	50,000	1356	0.02712	[0.0257, 0.0286]	585,514	1.92x
PyMatching v2 (C++)	5	100,000	1596	0.01596	[0.0152, 0.0168]	276,643	1.00x
QECTOR CUDA Batch (GPU, unweighted)	5	100,000	6094	0.06094	[0.0595, 0.0624]	157,099	0.57x
QECTOR CUDA Batch (GPU, weighted)	5	100,000	3182	0.03182	[0.0307, 0.0329]	70,473	0.26x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR OpenCL Batch (GPU, unweighted)	5	100,000	6094	0.06094	[0.0595, 0.0624]	138,965	0.50x
QECTOR OpenCL Batch (GPU, weighted)	5	100,000	3182	0.03182	[0.0307, 0.0329]	51,241	0.18x
QECTOR Sparse Blossom (CPU)	5	100,000	1596	0.01596	[0.0152, 0.0168]	102,728	0.37x
QECTOR Union-Find (CPU)	5	100,000	2645	0.02645	[0.0255, 0.0275]	630,493	2.28x
PyMatching v2 (C++)	7	1,000	11	0.01100	[0.0062, 0.0196]	90,950	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	1,000	44	0.04400	[0.0329, 0.0586]	20,548	0.23x
QECTOR CUDA Batch (GPU, weighted)	7	1,000	25	0.02500	[0.0170, 0.0366]	4,149	0.05x
QECTOR OpenCL Batch (GPU, unweighted)	7	1,000	44	0.04400	[0.0329, 0.0586]	40,397	0.44x
QECTOR OpenCL Batch (GPU, weighted)	7	1,000	25	0.02500	[0.0170, 0.0366]	4,692	0.05x
QECTOR Sparse Blossom (CPU)	7	1,000	11	0.01100	[0.0062, 0.0196]	17,370	0.19x
QECTOR Union-Find (CPU)	7	1,000	18	0.01800	[0.0114, 0.0283]	70,581	0.78x
PyMatching v2 (C++)	7	5,000	74	0.01480	[0.0118, 0.0185]	97,446	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	5,000	218	0.04360	[0.0383, 0.0496]	57,195	0.59x
QECTOR CUDA Batch (GPU, weighted)	7	5,000	105	0.02100	[0.0174, 0.0254]	14,833	0.15x
QECTOR OpenCL Batch (GPU, unweighted)	7	5,000	218	0.04360	[0.0383, 0.0496]	57,890	0.59x
QECTOR OpenCL Batch (GPU, weighted)	7	5,000	105	0.02100	[0.0174, 0.0254]	9,637	0.10x
QECTOR Sparse Blossom (CPU)	7	5,000	75	0.01500	[0.0120, 0.0188]	17,735	0.18x
QECTOR Union-Find (CPU)	7	5,000	107	0.02140	[0.0177, 0.0258]	142,737	1.47x
PyMatching v2 (C++)	7	10,000	130	0.01300	[0.0110, 0.0154]	100,602	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	10,000	409	0.04090	[0.0372, 0.0450]	56,319	0.56x
QECTOR CUDA Batch (GPU, weighted)	7	10,000	180	0.01800	[0.0156, 0.0208]	12,680	0.13x
QECTOR OpenCL Batch (GPU, unweighted)	7	10,000	409	0.04090	[0.0372, 0.0450]	50,970	0.51x
QECTOR OpenCL Batch (GPU, weighted)	7	10,000	180	0.01800	[0.0156, 0.0208]	8,063	0.08x
QECTOR Sparse Blossom (CPU)	7	10,000	133	0.01330	[0.0112, 0.0157]	17,979	0.18x
QECTOR Union-Find (CPU)	7	10,000	209	0.02090	[0.0183, 0.0239]	133,876	1.33x
PyMatching v2 (C++)	7	50,000	619	0.01238	[0.0114, 0.0134]	100,970	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	50,000	2197	0.04394	[0.0422, 0.0458]	40,031	0.40x
QECTOR OpenCL Batch (GPU, unweighted)	7	50,000	2197	0.04394	[0.0422, 0.0458]	223,597	2.21x
QECTOR Sparse Blossom (CPU)	7	50,000	624	0.01248	[0.0115, 0.0135]	17,994	0.18x
QECTOR Union-Find (CPU)	7	50,000	1055	0.02110	[0.0199, 0.0224]	150,367	1.49x
PyMatching v2 (C++)	7	100,000	1220	0.01220	[0.0115, 0.0129]	101,498	1.00x
QECTOR CUDA Batch (GPU, unweighted)	7	100,000	4274	0.04274	[0.0415, 0.0440]	42,492	0.42x
QECTOR OpenCL Batch (GPU, unweighted)	7	100,000	4274	0.04274	[0.0415, 0.0440]	237,312	2.34x
QECTOR Sparse Blossom (CPU)	7	100,000	1233	0.01233	[0.0117, 0.0130]	17,018	0.17x
QECTOR Union-Find (CPU)	7	100,000	2042	0.02042	[0.0196, 0.0213]	152,985	1.51x
PyMatching v2 (C++)	9	1,000	7	0.00700	[0.0034, 0.0144]	35,063	1.00x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR CUDA Batch (GPU, unweighted)	9	1,000	56	0.05600	[0.0434, 0.0720]	8,592	0.24x
QECTOR CUDA Batch (GPU, weighted)	9	1,000	17	0.01700	[0.0106, 0.0271]	924	0.03x
QECTOR OpenCL Batch (GPU, unweighted)	9	1,000	56	0.05600	[0.0434, 0.0720]	15,119	0.43x
QECTOR OpenCL Batch (GPU, weighted)	9	1,000	17	0.01700	[0.0106, 0.0271]	1,000	0.03x
QECTOR Sparse Blossom (CPU)	9	1,000	7	0.00700	[0.0034, 0.0144]	2,322	0.07x
QECTOR Union-Find (CPU)	9	1,000	14	0.01400	[0.0084, 0.0234]	17,950	0.51x
PyMatching v2 (C++)	9	5,000	36	0.00720	[0.0052, 0.0100]	39,197	1.00x
QECTOR CUDA Batch (GPU, unweighted)	9	5,000	224	0.04480	[0.0394, 0.0509]	28,984	0.74x
QECTOR CUDA Batch (GPU, weighted)	9	5,000	57	0.01140	[0.0088, 0.0147]	3,765	0.10x
QECTOR OpenCL Batch (GPU, unweighted)	9	5,000	224	0.04480	[0.0394, 0.0509]	22,434	0.57x
QECTOR OpenCL Batch (GPU, weighted)	9	5,000	57	0.01140	[0.0088, 0.0147]	1,984	0.05x
QECTOR Sparse Blossom (CPU)	9	5,000	36	0.00720	[0.0052, 0.0100]	2,569	0.07x
QECTOR Union-Find (CPU)	9	5,000	85	0.01700	[0.0138, 0.0210]	44,132	1.13x
PyMatching v2 (C++)	9	10,000	78	0.00780	[0.0063, 0.0097]	40,425	1.00x
QECTOR CUDA Batch (GPU, unweighted)	9	10,000	431	0.04310	[0.0393, 0.0473]	24,070	0.59x
QECTOR OpenCL Batch (GPU, unweighted)	9	10,000	431	0.04310	[0.0393, 0.0473]	18,874	0.47x
QECTOR Sparse Blossom (CPU)	9	10,000	79	0.00790	[0.0063, 0.0098]	2,569	0.06x
QECTOR Union-Find (CPU)	9	10,000	156	0.01560	[0.0134, 0.0182]	37,363	0.92x
PyMatching v2 (C++)	9	50,000	450	0.00900	[0.0082, 0.0099]	39,995	1.00x
QECTOR CUDA Batch (GPU, unweighted)	9	50,000	2324	0.04648	[0.0447, 0.0484]	15,442	0.39x
QECTOR OpenCL Batch (GPU, unweighted)	9	50,000	2324	0.04648	[0.0447, 0.0484]	101,861	2.55x
QECTOR Sparse Blossom (CPU)	9	50,000	453	0.00906	[0.0083, 0.0099]	2,343	0.06x
QECTOR Union-Find (CPU)	9	50,000	827	0.01654	[0.0155, 0.0177]	40,492	1.01x
PyMatching v2 (C++)	9	100,000	878	0.00878	[0.0082, 0.0094]	40,581	1.00x
QECTOR OpenCL Batch (GPU, unweighted)	9	100,000	4663	0.04663	[0.0453, 0.0480]	101,976	2.51x
QECTOR Union-Find (CPU)	9	100,000	1732	0.01732	[0.0165, 0.0181]	42,073	1.04x
PyMatching v2 (C++)	11	1,000	7	0.00700	[0.0034, 0.0144]	18,584	1.00x
QECTOR CUDA Batch (GPU, unweighted)	11	1,000	44	0.04400	[0.0329, 0.0586]	5,091	0.27x
QECTOR CUDA Batch (GPU, weighted)	11	1,000	10	0.01000	[0.0054, 0.0183]	277	0.01x
QECTOR OpenCL Batch (GPU, unweighted)	11	1,000	44	0.04400	[0.0329, 0.0586]	7,160	0.39x
QECTOR OpenCL Batch (GPU, weighted)	11	1,000	10	0.01000	[0.0054, 0.0183]	286	0.01x
QECTOR Sparse Blossom (CPU)	11	1,000	7	0.00700	[0.0034, 0.0144]	701	0.04x
QECTOR Union-Find (CPU)	11	1,000	13	0.01300	[0.0076, 0.0221]	4,544	0.24x
PyMatching v2 (C++)	11	5,000	24	0.00480	[0.0032, 0.0071]	20,329	1.00x
QECTOR CUDA Batch (GPU, unweighted)	11	5,000	229	0.04580	[0.0403, 0.0520]	14,373	0.71x
QECTOR OpenCL Batch (GPU, unweighted)	11	5,000	229	0.04580	[0.0403, 0.0520]	10,941	0.54x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
QECTOR Sparse Blossom (CPU)	11	5,000	24	0.00480	[0.0032, 0.0071]	721	0.04x
QECTOR Union-Find (CPU)	11	5,000	71	0.01420	[0.0113, 0.0179]	8,186	0.40x
PyMatching v2 (C++)	11	10,000	77	0.00770	[0.0062, 0.0096]	19,444	1.00x
QECTOR CUDA Batch (GPU, unweighted)	11	10,000	440	0.04400	[0.0402, 0.0482]	11,862	0.61x
QECTOR OpenCL Batch (GPU, unweighted)	11	10,000	440	0.04400	[0.0402, 0.0482]	9,328	0.48x
QECTOR Sparse Blossom (CPU)	11	10,000	78	0.00780	[0.0063, 0.0097]	757	0.04x
QECTOR Union-Find (CPU)	11	10,000	172	0.01720	[0.0148, 0.0199]	8,019	0.41x
PyMatching v2 (C++)	11	50,000	334	0.00668	[0.0060, 0.0074]	18,836	1.00x
QECTOR OpenCL Batch (GPU, unweighted)	11	50,000	2111	0.04222	[0.0405, 0.0440]	50,184	2.66x
QECTOR Union-Find (CPU)	11	50,000	839	0.01678	[0.0157, 0.0179]	8,681	0.46x
PyMatching v2 (C++)	11	100,000	647	0.00647	[0.0060, 0.0070]	19,880	1.00x
PyMatching v2 (C++)	13	1,000	3	0.00300	[0.0010, 0.0088]	7,679	1.00x
QECTOR CUDA Batch (GPU, unweighted)	13	1,000	39	0.03900	[0.0287, 0.0529]	2,747	0.36x
QECTOR CUDA Batch (GPU, weighted)	13	1,000	8	0.00800	[0.0041, 0.0157]	123	0.02x
QECTOR OpenCL Batch (GPU, unweighted)	13	1,000	39	0.03900	[0.0287, 0.0529]	4,002	0.52x
QECTOR OpenCL Batch (GPU, weighted)	13	1,000	8	0.00800	[0.0041, 0.0157]	137	0.02x
QECTOR Sparse Blossom (CPU)	13	1,000	3	0.00300	[0.0010, 0.0088]	220	0.03x
QECTOR Union-Find (CPU)	13	1,000	11	0.01100	[0.0062, 0.0196]	1,066	0.14x
PyMatching v2 (C++)	13	5,000	18	0.00360	[0.0023, 0.0057]	9,295	1.00x
QECTOR CUDA Batch (GPU, unweighted)	13	5,000	201	0.04020	[0.0351, 0.0460]	7,927	0.85x
QECTOR OpenCL Batch (GPU, unweighted)	13	5,000	201	0.04020	[0.0351, 0.0460]	6,134	0.66x
QECTOR Sparse Blossom (CPU)	13	5,000	19	0.00380	[0.0024, 0.0059]	227	0.02x
QECTOR Union-Find (CPU)	13	5,000	98	0.01960	[0.0161, 0.0238]	1,623	0.17x
PyMatching v2 (C++)	13	10,000	37	0.00370	[0.0027, 0.0051]	7,786	1.00x
QECTOR CUDA Batch (GPU, unweighted)	13	10,000	410	0.04100	[0.0373, 0.0451]	6,600	0.85x
QECTOR OpenCL Batch (GPU, unweighted)	13	10,000	410	0.04100	[0.0373, 0.0451]	24,690	3.17x
QECTOR Union-Find (CPU)	13	10,000	140	0.01400	[0.0119, 0.0165]	2,379	0.30x
PyMatching v2 (C++)	13	50,000	221	0.00442	[0.0039, 0.0050]	7,769	1.00x
PyMatching v2 (C++)	13	100,000	445	0.00445	[0.0041, 0.0049]	10,008	1.00x
PyMatching v2 (C++)	15	1,000	2	0.00200	[0.0005, 0.0073]	5,999	1.00x
QECTOR CUDA Batch (GPU, unweighted)	15	1,000	41	0.04100	[0.0304, 0.0551]	1,654	0.28x
QECTOR OpenCL Batch (GPU, unweighted)	15	1,000	41	0.04100	[0.0304, 0.0551]	2,412	0.40x
QECTOR Sparse Blossom (CPU)	15	1,000	2	0.00200	[0.0005, 0.0073]	59	0.01x
QECTOR Union-Find (CPU)	15	1,000	12	0.01200	[0.0069, 0.0209]	317	0.05x
PyMatching v2 (C++)	15	5,000	12	0.00240	[0.0014, 0.0042]	5,784	1.00x
QECTOR CUDA Batch (GPU, unweighted)	15	5,000	170	0.03400	[0.0293, 0.0394]	4,780	0.83x
QECTOR OpenCL Batch (GPU, unweighted)	15	5,000	170	0.03400	[0.0293, 0.0394]	12,703	2.20x

Decoder	d	Shots	Errors	LER	95% CI	dec/s	vs PM
PyMatching v2 (C++)	15	10,000	44	0.00440	[0.0033, 0.0059]	6,863	1.00x
QECTOR CUDA Batch (GPU, unweighted)	15	10,000	376	0.03760	[0.0340, 0.0415]	3,971	0.58x
QECTOR OpenCL Batch (GPU, unweighted)	15	10,000	376	0.03760	[0.0340, 0.0415]	11,531	1.68x
PyMatching v2 (C++)	15	50,000	171	0.00342	[0.0029, 0.0040]	6,773	1.00x
PyMatching v2 (C++)	15	100,000	314	0.00314	[0.0028, 0.0035]	6,721	1.00x

Not measured (95 cells)

Listed so the gaps are explicit. These cells were never run and carry no numbers; nothing here was extrapolated or synthesised.

Decoder	d	Shots	Reason	Projected
ldpc BP-OSD	3	100,000	over per-cell decode budget	40s
ldpc BP-OSD	5	5,000	over per-cell decode budget	48s
ldpc BP-OSD	5	10,000	over per-cell decode budget	96s
ldpc BP-OSD	5	50,000	over per-cell decode budget	480s
ldpc BP-OSD	5	100,000	over per-cell decode budget	960s
QECTOR CUDA Batch (GPU, weighted)	7	50,000	over per-cell decode budget	42s
QECTOR CUDA Batch (GPU, weighted)	7	100,000	over per-cell decode budget	84s
QECTOR OpenCL Batch (GPU, weighted)	7	50,000	over per-cell decode budget	55s
QECTOR OpenCL Batch (GPU, weighted)	7	100,000	over per-cell decode budget	110s
ldpc BP-OSD	7	1,000	over per-cell decode budget	36s
ldpc BP-OSD	7	5,000	over per-cell decode budget	180s
ldpc BP-OSD	7	10,000	over per-cell decode budget	359s
ldpc BP-OSD	7	50,000	over per-cell decode budget	1,797s
ldpc BP-OSD	7	100,000	over per-cell decode budget	3,594s
QECTOR Sparse Blossom (CPU)	9	100,000	over per-cell decode budget	48s
QECTOR CUDA Batch (GPU, unweighted)	9	100,000	over per-cell decode budget	39s
QECTOR CUDA Batch (GPU, weighted)	9	10,000	over per-cell decode budget	30s
QECTOR CUDA Batch (GPU, weighted)	9	50,000	over per-cell decode budget	151s
QECTOR CUDA Batch (GPU, weighted)	9	100,000	over per-cell decode budget	302s
QECTOR OpenCL Batch (GPU, weighted)	9	10,000	over per-cell decode budget	30s
QECTOR OpenCL Batch (GPU, weighted)	9	50,000	over per-cell decode budget	152s
QECTOR OpenCL Batch (GPU, weighted)	9	100,000	over per-cell decode budget	304s
ldpc BP-OSD	9	1,000	over per-cell decode budget	99s
ldpc BP-OSD	9	5,000	over per-cell decode budget	496s
ldpc BP-OSD	9	10,000	over per-cell decode budget	993s
ldpc BP-OSD	9	50,000	over per-cell decode budget	4,963s
ldpc BP-OSD	9	100,000	over per-cell decode budget	9,926s
QECTOR Sparse Blossom (CPU)	11	50,000	over per-cell decode budget	70s
QECTOR Sparse Blossom (CPU)	11	100,000	over per-cell decode budget	140s
QECTOR Union-Find (CPU)	11	100,000	over per-cell decode budget	41s
QECTOR CUDA Batch (GPU, unweighted)	11	50,000	over per-cell decode budget	42s
QECTOR CUDA Batch (GPU, unweighted)	11	100,000	over per-cell decode budget	84s
QECTOR CUDA Batch (GPU, weighted)	11	5,000	over per-cell decode budget	49s
QECTOR CUDA Batch (GPU, weighted)	11	10,000	over per-cell decode budget	98s
QECTOR CUDA Batch (GPU, weighted)	11	50,000	over per-cell decode budget	490s
QECTOR CUDA Batch (GPU, weighted)	11	100,000	over per-cell decode budget	980s
QECTOR OpenCL Batch (GPU, unweighted)	11	100,000	over per-cell decode budget	45s
QECTOR OpenCL Batch (GPU, weighted)	11	5,000	over per-cell decode budget	47s
QECTOR OpenCL Batch (GPU, weighted)	11	10,000	over per-cell decode budget	94s
QECTOR OpenCL Batch (GPU, weighted)	11	50,000	over per-cell decode budget	471s
QECTOR OpenCL Batch (GPU, weighted)	11	100,000	over per-cell decode budget	942s
ldpc BP-OSD	11	1,000	over per-cell decode budget	210s

Decoder	d	Shots	Reason	Projected
ldpc BP-OSD	11	5,000	over per-cell decode budget	1,048s
ldpc BP-OSD	11	10,000	over per-cell decode budget	2,096s
ldpc BP-OSD	11	50,000	over per-cell decode budget	10,479s
ldpc BP-OSD	11	100,000	over per-cell decode budget	20,957s
QECTOR Sparse Blossom (CPU)	13	10,000	over per-cell decode budget	48s
QECTOR Sparse Blossom (CPU)	13	50,000	over per-cell decode budget	239s
QECTOR Sparse Blossom (CPU)	13	100,000	over per-cell decode budget	478s
QECTOR Union-Find (CPU)	13	50,000	over per-cell decode budget	78s
QECTOR Union-Find (CPU)	13	100,000	over per-cell decode budget	157s
QECTOR CUDA Batch (GPU, unweighted)	13	50,000	over per-cell decode budget	68s
QECTOR CUDA Batch (GPU, unweighted)	13	100,000	over per-cell decode budget	135s
QECTOR CUDA Batch (GPU, weighted)	13	5,000	over per-cell decode budget	101s
QECTOR CUDA Batch (GPU, weighted)	13	10,000	over per-cell decode budget	202s
QECTOR CUDA Batch (GPU, weighted)	13	50,000	over per-cell decode budget	1,009s
QECTOR CUDA Batch (GPU, weighted)	13	100,000	over per-cell decode budget	2,017s
QECTOR OpenCL Batch (GPU, unweighted)	13	50,000	over per-cell decode budget	41s
QECTOR OpenCL Batch (GPU, unweighted)	13	100,000	over per-cell decode budget	81s
QECTOR OpenCL Batch (GPU, weighted)	13	5,000	over per-cell decode budget	99s
QECTOR OpenCL Batch (GPU, weighted)	13	10,000	over per-cell decode budget	197s
QECTOR OpenCL Batch (GPU, weighted)	13	50,000	over per-cell decode budget	986s
QECTOR OpenCL Batch (GPU, weighted)	13	100,000	over per-cell decode budget	1,973s
ldpc BP-OSD	13	1,000	over per-cell decode budget	362s
ldpc BP-OSD	13	5,000	over per-cell decode budget	1,808s
ldpc BP-OSD	13	10,000	over per-cell decode budget	3,616s
ldpc BP-OSD	13	50,000	over per-cell decode budget	18,080s
ldpc BP-OSD	13	100,000	over per-cell decode budget	36,160s
QECTOR Sparse Blossom (CPU)	15	5,000	over per-cell decode budget	89s
QECTOR Sparse Blossom (CPU)	15	10,000	over per-cell decode budget	178s
QECTOR Sparse Blossom (CPU)	15	50,000	over per-cell decode budget	891s
QECTOR Sparse Blossom (CPU)	15	100,000	over per-cell decode budget	1,782s
QECTOR Union-Find (CPU)	15	5,000	over per-cell decode budget	26s
QECTOR Union-Find (CPU)	15	10,000	over per-cell decode budget	52s
QECTOR Union-Find (CPU)	15	50,000	over per-cell decode budget	262s
QECTOR Union-Find (CPU)	15	100,000	over per-cell decode budget	523s
QECTOR CUDA Batch (GPU, unweighted)	15	50,000	over per-cell decode budget	114s
QECTOR CUDA Batch (GPU, unweighted)	15	100,000	over per-cell decode budget	228s
QECTOR CUDA Batch (GPU, weighted)	15	1,000	over per-cell decode budget	44s
QECTOR CUDA Batch (GPU, weighted)	15	5,000	over per-cell decode budget	222s
QECTOR CUDA Batch (GPU, weighted)	15	10,000	over per-cell decode budget	445s
QECTOR CUDA Batch (GPU, weighted)	15	50,000	over per-cell decode budget	2,223s
QECTOR CUDA Batch (GPU, weighted)	15	100,000	over per-cell decode budget	4,446s
QECTOR OpenCL Batch (GPU, unweighted)	15	50,000	over per-cell decode budget	71s
QECTOR OpenCL Batch (GPU, unweighted)	15	100,000	over per-cell decode budget	142s

Decoder	d	Shots	Reason	Projected
QECTOR OpenCL Batch (GPU, weighted)	15	1,000	over per-cell decode budget	34s
QECTOR OpenCL Batch (GPU, weighted)	15	5,000	over per-cell decode budget	170s
QECTOR OpenCL Batch (GPU, weighted)	15	10,000	over per-cell decode budget	340s
QECTOR OpenCL Batch (GPU, weighted)	15	50,000	over per-cell decode budget	1,701s
QECTOR OpenCL Batch (GPU, weighted)	15	100,000	over per-cell decode budget	3,403s
ldpc BP-OSD	15	1,000	over per-cell decode budget	557s
ldpc BP-OSD	15	5,000	over per-cell decode budget	2,783s
ldpc BP-OSD	15	10,000	over per-cell decode budget	5,566s
ldpc BP-OSD	15	50,000	over per-cell decode budget	27,830s
ldpc BP-OSD	15	100,000	over per-cell decode budget	55,660s